

Pairs of $f(x)$ & $f'(x)$

$$\textcircled{1} \int f'(x) [f(x)]^n dx = \frac{[f(x)]^{n+1}}{n+1} + c, \quad \boxed{n \neq -1}$$

$$\textcircled{2} \int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\textcircled{3} \int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

" x "

$$\textcircled{1} \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c, \quad \boxed{\begin{matrix} n \neq -1 \\ a \neq 0 \end{matrix}}$$

$$\textcircled{2} \int \frac{1}{ax+b} dx = \frac{1}{a} \ln |ax+b| + c, \quad \boxed{a \neq 0}$$

$$\begin{aligned} &\text{If } a = 1, \\ &\Rightarrow \int \frac{1}{x+b} dx = \ln |x+b| + c \end{aligned}$$

" x^2 "

$$\textcircled{1} \int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

" e "

If $a=1, b=0$
 $\Rightarrow \int e^x dx = e^x$

$$\textcircled{1} \int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$a \neq 0$

$$\textcircled{2} \int f'(x) e^{f(x)} dx = e^{f(x)} + C$$

$\sin x$

$$\textcircled{1} \int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

If $a=1,$

$$\int \sin(x+b) dx = -\cos(x+b) + C$$

$a \neq 0$

$\cos x$

$$\textcircled{1} \int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

If $a = 1$,

$$\int \cos(x+b) dx = \sin(x+b) + C$$

$a \neq 0$

$\sec^2 x$

$$\textcircled{1} \int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

If $a = 1$,

$$\int \sec^2(x+b) dx = \tan(x+b) + C$$

$a \neq 0$

Pairs of $f(x)$ & $f'(x)$

$$\textcircled{1} \int f'(x) \sin f(x) dx = -\cos f(x) + C$$

$$\textcircled{2} \int f'(x) \cos f(x) dx = \sin f(x) + C$$

$$\textcircled{3} \int f'(x) \sec^2 f(x) dx = \tan f(x) + C$$

Common Tricks

$$\int \sin^2 x$$

$$\begin{aligned} & \int \sin^2 x \, dx \\ &= \int \frac{1}{2} (1 - \cos 2x) \, dx \\ &= \frac{1}{2} \int (1 - \cos 2x) \, dx \\ &= \frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right) + C \\ &= \frac{1}{2} x - \frac{1}{4} \sin 2x + C \end{aligned}$$

We know that
 $\cos 2x = 1 - 2\sin^2 x$
 $\Rightarrow \sin^2 x = \frac{1}{2} (1 - \cos 2x)$

$$\int \cos^2 x$$

$$\begin{aligned} & \int \cos^2 x \, dx \\ &= \int \frac{1}{2} (1 + \cos 2x) \, dx \\ &= \frac{1}{2} \int (1 + \cos 2x) \, dx \\ &= \frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right) + C \\ &= \frac{1}{2} x + \frac{1}{4} \sin 2x + C \end{aligned}$$

We know that
 $\cos 2x = 2\cos^2 x - 1$
 $\Rightarrow \cos^2 x = \frac{1}{2} (1 + \cos 2x)$

$$\text{let } u = \cos x \\ du = -\sin x \, dx$$

$$\int \sin^3 x$$

$$\begin{aligned} & \int \sin^3 x \, dx \\ &= \int (\sin^2 x)(\sin x) \, dx \\ &= \int (1 - \cos^2 x)(\sin x) \, dx \\ &= \int \sin x + (-\sin x)(\cos x)^2 \, dx \\ &= -\cos x + \frac{(\cos x)^3}{3} + C \\ &= \frac{1}{3} \cos^3 x - \cos x + C \quad \# \end{aligned}$$

OR

$$\text{We know that} \\ \sin^2 x + \cos^2 x = 1 \\ \Rightarrow \sin^2 x = 1 - \cos^2 x$$

$$\begin{aligned} &= - \int (1 - u^2) \, du \\ &= - \left[u - \frac{u^3}{3} \right] \\ &= -u + \frac{u^3}{3} \\ &= -\cos x + \frac{\cos^3 x}{3} \\ &= \frac{1}{3} \cos^3 x - \cos x + C \quad \# \end{aligned}$$

$$\int \cos^3 x$$

$$\begin{aligned} & \int \cos^3 x \, dx \\ &= \int (\cos^2 x)(\cos x) \, dx \\ &= \int (1 - \sin^2 x)(\cos x) \, dx \\ &= \int \cos x - (\cos x)(\sin x)^2 \, dx \\ &= \sin x - \frac{(\sin x)^3}{3} \\ &= -\frac{1}{3} \sin^3 x + \sin x + C \quad \# \end{aligned}$$

OR

$$\text{We know that} \\ \sin^2 x + \cos^2 x = 1 \\ \Rightarrow \cos^2 x = 1 - \sin^2 x$$

$$\text{let } u = \sin x \\ du = \cos x \, dx$$

$$\begin{aligned} &= \int (1 - u^2) \, du \\ &= u - \frac{u^3}{3} + C \\ &= \sin x - \frac{\sin^3 x}{3} + C \\ &= -\frac{1}{3} \sin^3 x + \sin x + C \end{aligned}$$

$$\int \sin^4 x$$

$$\begin{aligned} \int \sin^4 x \, dx &= \int [\sin^2 x]^2 \, dx \\ &= \int \left[\frac{1}{2} (1 - \cos 2x) \right]^2 \, dx \\ &= \frac{1}{4} \int (1 - \cos 2x)^2 \, dx \\ &= \frac{1}{4} \int [1 - 2\cos 2x + \cos^2 2x] \, dx \end{aligned}$$

$$\begin{aligned} &\rightarrow = \frac{1}{4} \int \left[1 - 2\cos 2x + \frac{1}{2} (1 + \cos 4x) \right] \, dx \\ &= \frac{1}{4} \int \left[\frac{3}{2} - 2\cos 2x + \frac{1}{2} \cos 4x \right] \, dx \\ &= \frac{1}{4} \left(\frac{3}{2}x - \frac{2}{2} \sin 2x - \frac{1}{8} \sin 4x \right) \\ &= \frac{3}{8}x - \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x + C \quad \# \end{aligned}$$

$$\int \cos^4 x$$

$$\begin{aligned} \int \cos^4 x \, dx &= \int [\cos^2 x]^2 \, dx \\ &= \int \left[\frac{1}{2} (1 + \cos 2x) \right]^2 \, dx \\ &= \frac{1}{4} \int (1 + \cos 2x)^2 \, dx \\ &= \frac{1}{4} \int [1 + 2\cos 2x + \cos^2 2x] \, dx \end{aligned}$$

$$\begin{aligned} &\rightarrow \frac{1}{4} \int [1 + 2\cos 2x + \frac{1}{2} (1 + \cos 4x)] \, dx \\ &= \frac{1}{4} \int [1 + 2\cos 2x + \frac{1}{2} + \frac{1}{2} \cos 4x] \, dx \\ &= \frac{1}{4} \int \left[\frac{3}{2} + 2\cos 2x + \frac{1}{2} \cos 4x \right] \, dx \\ &= \frac{1}{4} \left[\frac{3}{2}x + \sin 2x + \frac{1}{8} \sin 4x \right] \\ &= \frac{3}{8}x + \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x + C \quad \# \end{aligned}$$

$$\int \tan^2 x$$

$$\begin{aligned} & \int \tan^2 x \, dx \\ &= \int (\sec^2 x - 1) \, dx \\ &= \tan x - x + C \end{aligned}$$

$$\begin{aligned} & \text{We know that} \\ & 1 + \tan^2 x = \sec^2 x \\ & \Rightarrow \tan^2 x = \sec^2 x - 1 \end{aligned}$$

$$\int \tan x$$

$$\begin{aligned} & \int \tan x \, dx \\ &= \int \frac{\sin x}{\cos x} \, dx \\ &= - \int \frac{-\sin x}{\cos x} \\ &= -\ln |\cos x| + C \end{aligned}$$

$$\int \frac{f'(x)}{f(x)} = \ln |f(x)| + C$$

$$\int \cot x$$

$$\int \cot x \, dx$$

$$= \int \frac{1}{\tan x} \, dx$$

$$= \int \frac{\cos x}{\sin x} \, dx$$

$$= \ln |\sin x| + C$$

$$\int \frac{f'(x)}{f(x)} = \ln |f(x)| + C$$

Rational Fractions

Improper $\Rightarrow \frac{f(x)}{g(x)}$, $\deg f(x) \geq \deg g(x)$

[Do long division first]

Example 1:

$$\int \frac{x^2 + 1}{2 - 3x} dx$$

$$\begin{array}{r} -\frac{1}{3}x - \frac{2}{9} \\ -3x + 2 \overline{) x^2 + 0x + 1} \\ \underline{-x^2 - \frac{2}{3}x} \\ \frac{2}{3}x + 1 \\ \underline{\frac{2}{3}x - \frac{4}{9}} \\ \frac{13}{9} \end{array}$$

$$\therefore \frac{x^2 + 1}{2 - 3x} = \left(-\frac{1}{3}x - \frac{2}{9}\right) + \frac{13/9}{-3x + 2}$$

$$\begin{aligned} \int \frac{x^2 + 1}{2 - 3x} &= \int \left[-\frac{1}{3}x - \frac{2}{9} + \frac{13/9}{-3x + 2} \right] dx \\ &= -\frac{1}{6}x^2 - \frac{2}{9}x + \frac{13}{9} \left(-\frac{1}{3}\right) \ln|-3x + 2| + C \end{aligned}$$

Proper $\Rightarrow \frac{f(x)}{g(x)}$, $\deg f(x) < \deg g(x)$

[No partial fraction]

Example 1:

$$\int \frac{2x}{(x-1)^2(x+2)} dx$$

$$\text{let } \frac{2x}{(x-1)^2(x+2)} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x+2)}$$

$$2x = A(x-1)(x+2) + B(x+2) + C(x-1)^2$$

$$\text{let } x = 1 \rightarrow 2 = B(1+2)$$

$$B = \frac{2}{3}$$

$$\text{let } x = -2 \rightarrow -4 = C(-2-1)^2$$

$$C = -\frac{4}{9}$$

$$\text{let } x = 0 \rightarrow 0 = A(-1)(2) + B(2) + C(-1)^2$$

$$0 = -2A + \left(\frac{2}{3}\right)(2) + \left(-\frac{4}{9}\right)(1)$$

$$A = \frac{4}{9}$$

$$\therefore \frac{2x}{(x-1)^2(x+2)} = \frac{4/9}{(x-1)} + \frac{2/3}{(x-1)^2} - \frac{4/9}{(x+2)}$$

$$\begin{aligned} \therefore \int \frac{2x}{(x-1)^2(x+2)} &= \int \left[\frac{4/9}{(x-1)} + \frac{2/3}{(x-1)^2} - \frac{4/9}{(x+2)} \right] dx \\ &= \int \left[\frac{4/9}{(x-1)} + \frac{2}{3}(x-1)^{-2} - \frac{4/9}{(x+2)} \right] dx \end{aligned}$$

cont. below



$$\begin{aligned}
&= \frac{4}{9} \ln|x-1| + \frac{2}{3} \frac{(x-1)^{-1}}{-1(1)} - \frac{4}{9} \ln|x+2| + C \\
&= \frac{4}{9} \ln|x-1| - \frac{2}{3(x-1)} - \frac{4}{9} \ln|x+2| + C \\
&= \frac{4}{9} [\ln|x-1| - \ln|x+2|] - \frac{2}{3(x-1)} + C \quad \#
\end{aligned}$$

Irreducible denominator

$$\Rightarrow b^2 - 4ac < 0$$

[Complete the squares]

Example 1:

$$\int \frac{1}{x^2 + 4x + 7} dx$$

$$= \int \frac{1}{(x+2)^2 + 3} dx$$

$$\begin{aligned}
&x^2 + 4x + 7 \\
&= x^2 + 4x + \left(\frac{4}{2(1)}\right)^2 - \left(\frac{4}{2(1)}\right)^2 + 7 \\
&= x^2 + 4x + (2)^2 - (2)^2 + 7 \\
&= (x+2)^2 + 3
\end{aligned}$$

$$\text{Let } u = x+2$$

$$\frac{du}{dx} = 1$$

$$du = dx$$

$$= \int \frac{1}{u^2 + (\sqrt{3})^2} du$$

$$= \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{u}{\sqrt{3}} \right) + C$$

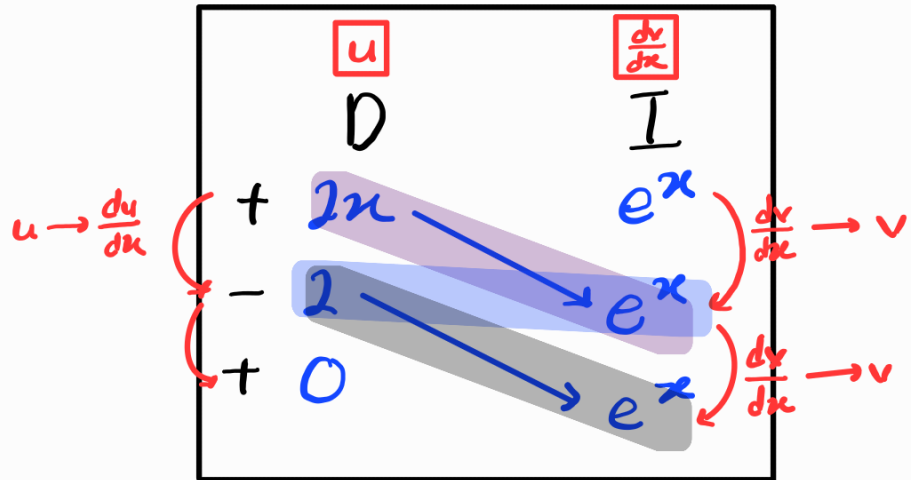
$$= \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x+2}{\sqrt{3}} \right) + C \quad \#$$

Integration by parts

Stop 1: When u becomes 0

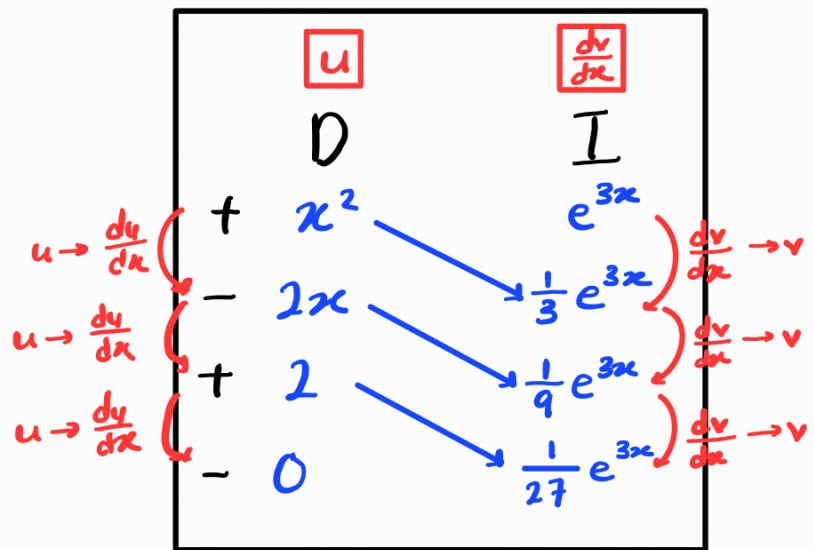
Example 1

$$\begin{aligned} & \int 1x e^x dx \\ &= 1xe^x - \int 1e^x \\ &= 1xe^x - 1e^x \end{aligned}$$



Example 2

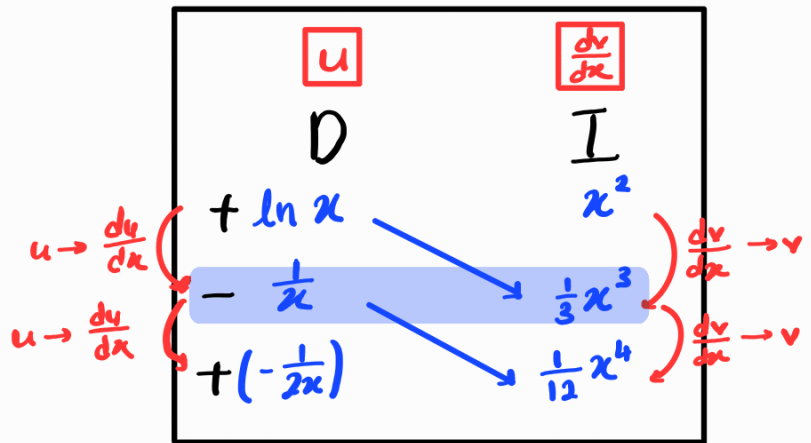
$$\begin{aligned} & \int x^2 e^{3x} dx \\ &= \frac{1}{3} x^2 e^{3x} - \int \frac{1}{3} x e^{3x} \\ &= \frac{1}{3} x^2 e^{3x} - \left[\frac{2}{9} x e^{3x} - \int \frac{2}{9} e^{3x} \right] \\ &= \frac{1}{3} x^2 e^{3x} - \frac{2}{9} x e^{3x} + \int \frac{2}{9} e^{3x} \\ &= \frac{1}{3} x^2 e^{3x} - \frac{2}{9} x e^{3x} + \frac{2}{27} e^{3x} + C \quad \# \end{aligned}$$



Stop 2: When row can be integrated

Example 1 :

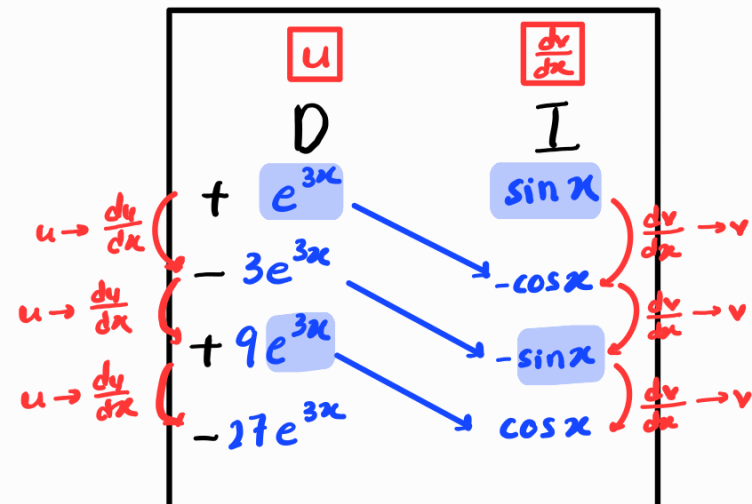
$$\begin{aligned} & \int x^2 \ln x \, dx \\ &= \frac{1}{3} x^3 \ln x - \int \left(\frac{1}{x} \right) \left(\frac{x^3}{3} \right) dx \\ &= \frac{1}{3} x^3 \ln x - \int \frac{x^2}{3} dx \\ &= \frac{1}{3} x^3 \ln x - \frac{x^3}{9} + C \, dx \end{aligned}$$



Stop 3: When row same as original

Example 1:

$$\int e^{3x} \sin x \, dx$$



$$\int e^{3x} \sin x = -e^{3x} \cos x - \int -3e^{3x} \cos x \, dx$$

$$\int e^{3x} \sin x = -e^{3x} \cos x + 3e^{3x} \sin x + \int -9e^{3x} \sin x \, dx$$

$$\int e^{3x} \sin x = -e^{3x} \cos x + 3e^{3x} \sin x - 9 \int e^{3x} \sin x \, dx$$

$$10 \int e^{3x} \sin x = -e^{3x} \cos x + 3e^{3x} \sin x + C$$

$$\int e^{3x} \sin x = -\frac{1}{10} e^{3x} \cos x + \frac{3}{10} e^{3x} \sin x + \frac{C}{10}$$

$$= -\frac{1}{10} e^{3x} \cos x + \frac{3}{10} e^{3x} \sin x + A, \text{ where } A = \frac{C}{10}$$

